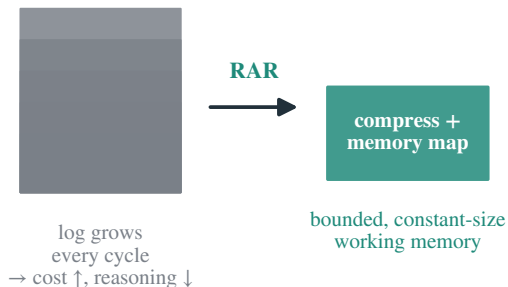


Recursive Autonomous Research (RAR): token-bounded agent memory

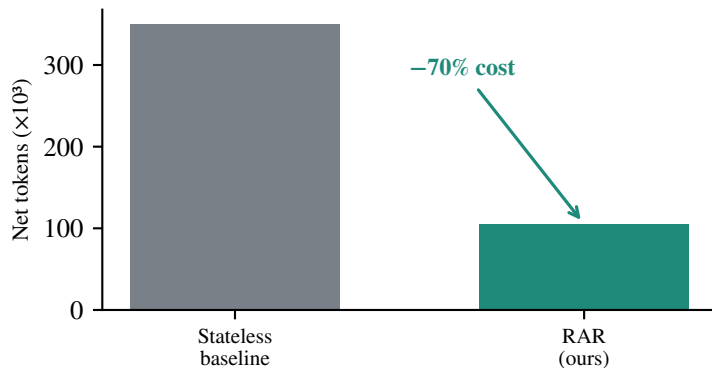
Keep a long-running AI agent affordable by compressing its own history as it works — same accuracy, a third of the cost.

The problem: “context rot”

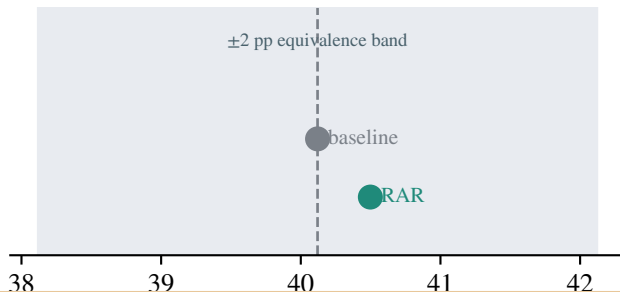


An AI research agent normally keeps a running log of everything it has tried. RAR periodically summarises it and stores past results in a small graph — so memory stays small instead of ballooning.

Validated: ~70% cheaper



...at statistically equal accuracy



Honest scope: efficiency is vs. other LLM agents (not classical optimisation, which uses 0 tokens). A proposed coherence metric gave a reported **NEGATIVE** result — diagnosed, not hidden.